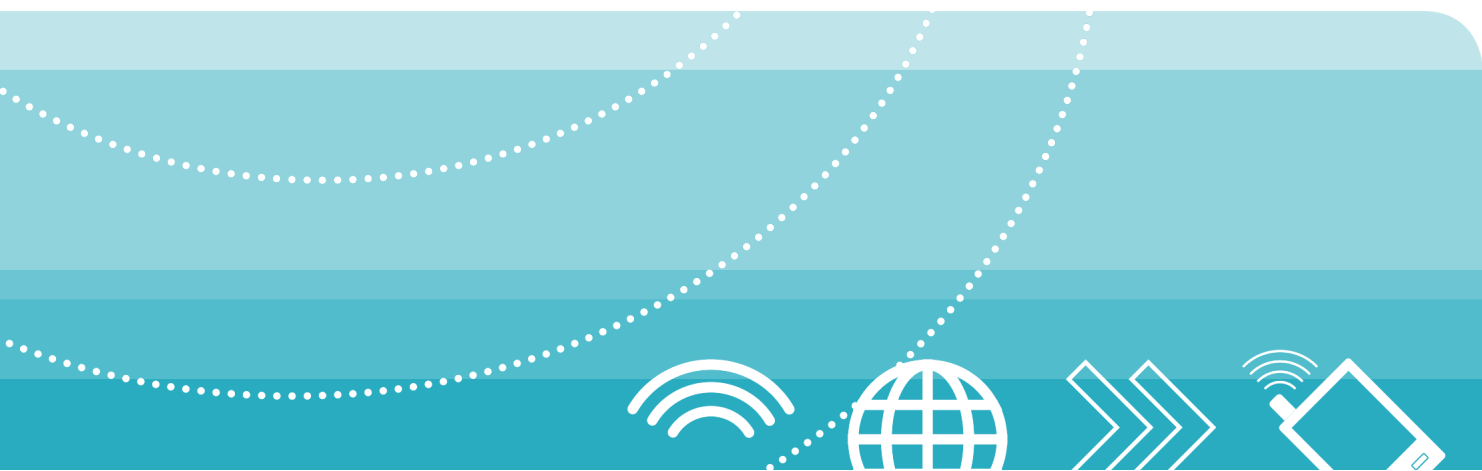




MT5710-CN 5G RedCap 模组 **Android 内核驱动集成指导**

V100R001C01

文档版本：01

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1 关于本文档

修订记录

文档版本	发布日期	说明
01	2023-06-09	第一次发布

2 目的

本文档主要是针对MT571X系列模组，基于Android系统的内核驱动集成开发活动进行相关的指导说明。

本文档主要面向基于Android系统的产品开发商的驱动开发人员。

3 范围

本文档适用于：

- 嵌入式Linux系统（Linux-4.4.1内核）及以上版本。
- Android 7.0（Linux-4.4.1内核）及以上版本的Android系统。
- 鼎桥 NR及LTE模块。

说明

如果不使用鼎桥模块的电源管理，则适用在内核版本2.6.18及以上的Linux系统上。

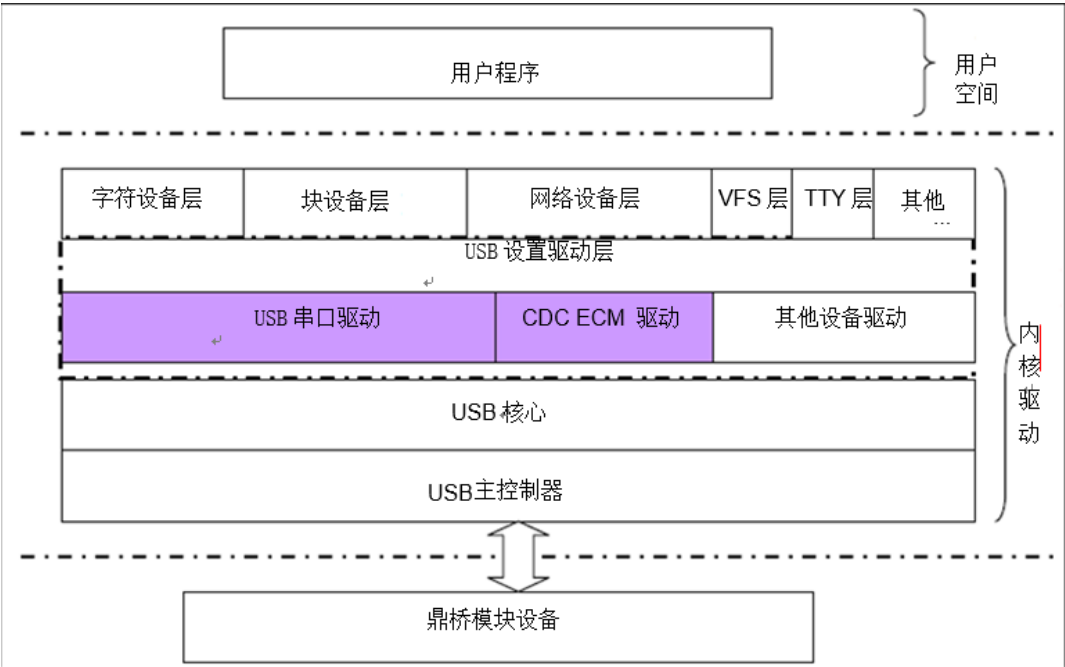
4 总体概述

4.1 Android系统的Linux内核支持鼎桥模块设备的驱动架构

4.1 Android 系统的 Linux 内核支持鼎桥模块设备的驱动架构

鼎桥MT571X设备和Android系统主要通过USB接口进行数据通信，Android系统上的Linux内核需要根据鼎桥模块设备上报的USB设备接口加载USB驱动，USB驱动正确加载后，鼎桥模块才能正常工作。

Android系统中支持鼎桥模块设备相关的Linux内核驱动架构，如下图所示：



如上图所示，在Linux系统中的USB驱动架构中，跟MT571X模块设备相关的驱动模块是USB串口驱动模块和CDC NCM驱动模块。

其中：

- **USB串口驱动**：USB的串口驱动，支持如AT端口等；该驱动模块的代码（usb-serial.c）已经内置在Linux内核的源码中。
- **CDC ECM驱动**：USB的标准ECM网口驱动，主要用于传输网络数据。

5 Android 系统的 Linux 内核驱动集成方案

[5.1 Android内核的USB串口驱动](#)

[5.2 Android内核的CDC NCM驱动](#)

[5.3 Android内核的CDC ECM驱动](#)

5.1 Android 内核的 USB 串口驱动

这部分集成涉及修改的Linux内核源码文件为：

Android_src/kernel/drivers/usb/serial/option.c

5.1.1 增加宏定义

步骤1：在Android_src/kernel/drivers/usb/serial/option.c文件中添加如下宏

```
#define TDTECH_VENDOR_ID 0x3466
```

步骤2：在Android_src/kernel/drivers/usb/serial/option.c文件中的static const struct usb_device_id option_ids[]的id列表中增加如下语句,用于匹配设备数据。

```
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x01, 0x03) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x01, 0x06) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x01, 0x0A) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x01, 0x0B) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x01, 0x0E) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x01, 0x10) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x01, 0x12) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x01, 0x13) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x01, 0x14) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x01, 0x15) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x01, 0x18) },
```

```
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x01, 0x19) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x01, 0x1A) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x01, 0x1B) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x01, 0x1C) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x01, 0x1D) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x02, 0x03) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x02, 0x06) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x02, 0x0A) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x02, 0x0B) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x02, 0x0E) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x02, 0x10) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x02, 0x12) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x02, 0x13) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x02, 0x14) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x02, 0x15) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x02, 0x18) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x02, 0x19) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x02, 0x1A) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x02, 0x1B) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x02, 0x1C) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x02, 0x1D) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x03, 0x03) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x03, 0x06) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x03, 0x0A) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x03, 0x0B) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x03, 0x0E) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x03, 0x10) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x03, 0x12) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x03, 0x13) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x03, 0x14) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x03, 0x15) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x03, 0x18) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x03, 0x19) },
```

```
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x03, 0x1A) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x03, 0x1B) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x03, 0x1C) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x03, 0x1D) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x04, 0x03) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x04, 0x06) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x04, 0x0A) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x04, 0x0B) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x04, 0x0E) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x04, 0x10) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x04, 0x12) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x04, 0x13) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x04, 0x14) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x04, 0x15) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x04, 0x18) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x04, 0x19) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x04, 0x1A) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x04, 0x1B) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x04, 0x1C) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x04, 0x1D) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x05, 0x03) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x05, 0x06) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x05, 0x0A) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x05, 0x0B) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x05, 0x0E) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x05, 0x10) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x05, 0x12) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x05, 0x13) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x05, 0x14) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x05, 0x15) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x05, 0x18) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x05, 0x19) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x05, 0x1A) },
```

```
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x05, 0x1B) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x05, 0x1C) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x05, 0x1D) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x06, 0x03) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x06, 0x06) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x06, 0x0A) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x06, 0x0B) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x06, 0x0E) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x06, 0x10) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x06, 0x12) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x06, 0x13) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x06, 0x14) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x06, 0x15) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x06, 0x18) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x06, 0x19) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x06, 0x1A) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x06, 0x1B) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x06, 0x1C) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x06, 0x1D) },
```

5.1.2 内核的编译配置

文本配置方式，修改Android_src目录下的.config文件。

- USB串口驱动相关的配置项：

CONFIG_USB_SERIAL=y

CONFIG_USB_SERIAL_OPTION=y

CONFIG_USB_SERIAL_WWAN=y

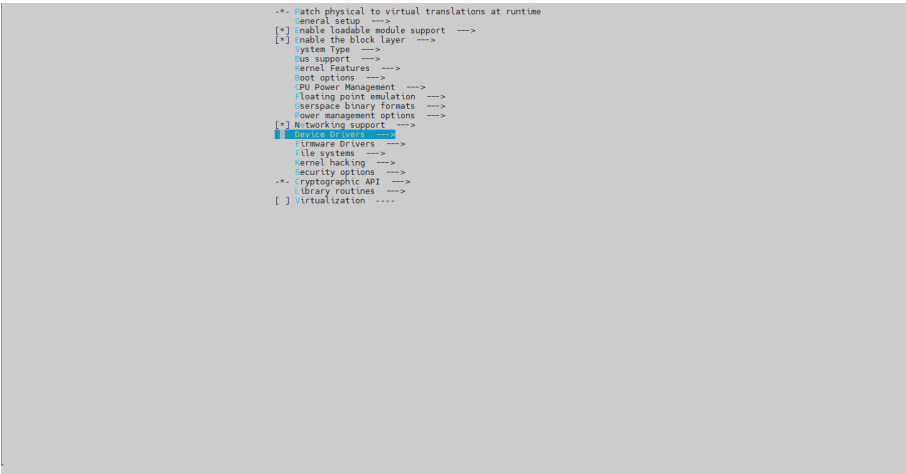
5.1.3 Make menuconfig 开启宏配置方式

Android_src/kernel/ 目录下执行命令 “make menuconfig”

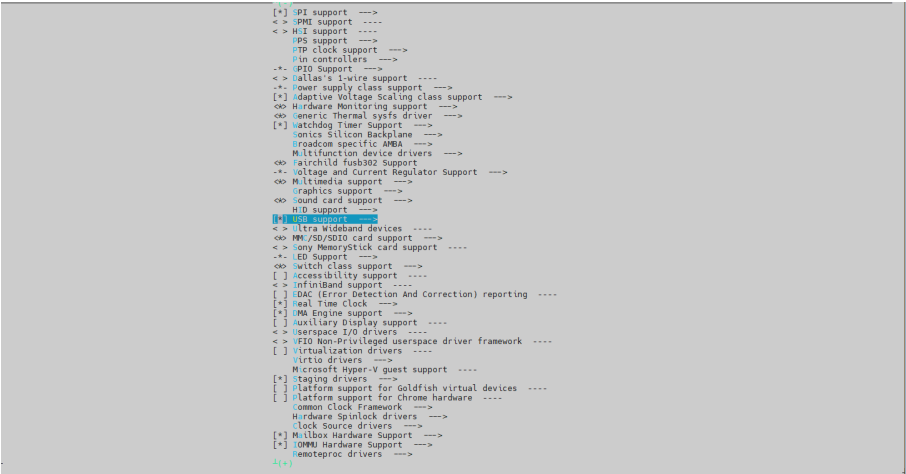
```
/Andriod9/code/kernel$ make menuconfig
```

出现Kernel Configuration界面后，进行如下配置：

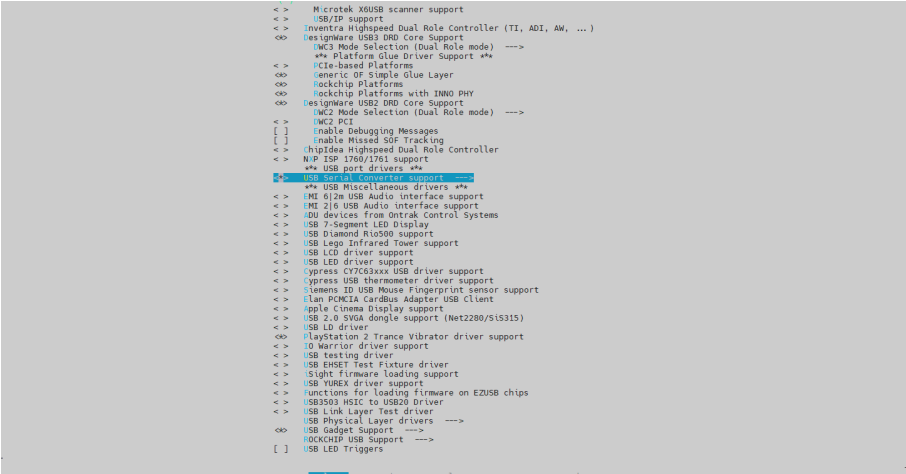
按Enter选择Device Drivers



按Enter选择USB support



按空格键选择USB Serial Converter support



保存退出，自动更新 .config 文件

5.2 Android 内核的 CDC NCM 驱动

5.2.1 Android 内核的编译配置

CDC NCM驱动的相关配置项，在Android_src根目录下的.config文件中，确保下面的配置项已经被选定：

CONFIG_USB_USBNET=y

CONFIG_NETDEVICES=y

CONFIG_USB_CONFIGFS_NCM=y

CONFIG_USB_FUNCTIONS_NCM=y

5.2.2 Make menuconfig 开启宏配置方式

Android_src/kernel/ 目录下执行命令 “make menuconfig”

```
/Andriod9/code/kernel$ make menuconfig
```

出现Kernel Configuration界面后，进行如下配置：

按Enter选择Device Drivers

```
-- match physical to virtual translations at runtime
General setup --->
[*] enable loadable module support --->
[*] enable the block layer --->
System type --->
Bus support --->
Kernel features --->
Root options --->
CPU Power Management --->
Floating point emulation --->
Userspace binary formats --->
Power management options --->
[*] Networking support --->
Device Drivers --->
  Firmware Drivers --->
  File systems --->
  Kernel hacking --->
  Security options --->
  Cryptographic API --->
  Library routines --->
  Virtualization --->
```

按Enter选择Network device support

```
[ ] enable AMB driver for NVIDIA Tegra SoCs
Generic Driver Options --->
Bus devices --->
< > connector - unified userspace <-> kernel space linker ----
< > Memory Technology Device (MTD) support ----
+ > Device Tree and Open Firmware support --->
< > Parallel port support ----
[*] Block devices --->
  SCSI devices --->
  M/M Express block device --->
  Misc devices --->
  ATA/ATAPI/IDE/RAID support (DEPRECATED) ----
  SCSI device support --->
  Serial ATA and Parallel ATA drivers (libata) --->
  Multiple devices driver support (RAID and LVM) --->
  Generic Target Core Mod (TCM) and ConfigFS Infrastructure ----
  Union MPT device support ----
  IEEE 1394 (FireWire) support --->
Network device support --->
  Open Channel SSD target support ----
  Input device support --->
  Character devices --->
  I2C support --->
  SPI support --->
  PPI support --->
  HSI support --->
  I2S support --->
  ITP clock support --->
  Pin controllers --->
  IPI support --->
  Dallas's I-wire support --->
  Power supply class support --->
  Adaptive Voltage Scaling class support --->
  Hardware Monitoring support --->
  Generic Thermal sysfs driver --->
  Watchdog Timer Support --->
  On-chip Silicon Backplane --->
  Broadcom specific MIB --->
  Multifunction device drivers --->
  Fathchild fusb302 support --->
  Voltage and Current Regulator Support --->
  Multimedia support --->
  Graphics support --->
  Sound card support --->
  HID support --->
  USB support --->
  Ultra Wideband devices --->
  MMC/SD/SDIO card support --->
  MemoryStick card support --->
  ...
```

按Enter选择USB Network Adapters

```

--- Network device support
[*] Network core driver support
< > bonding driver support
< > dummy net driver support
< > igmp (serial line load balancing) support
< > fibre Channel driver support
< > intermediate Functional Block support
< > ethernet team driver support
< > M-C-VLAN support
< > P-VLAN support
< > Virtual extensible Local Area Network (VXLAN)
< > generic Network Virtualization Encapsulation
< > Network console logging support
< > Universal TUN/TAP device driver support
[*] support for cross-endian vnet headers on little-endian kernels
< > virtual ethernet pair device
< > virtual netlink monitoring device
< > iRnet support
< > ** CAIF transport drivers **
distributed Switch Architecture drivers
[*] ethernet driver support
< > BDI driver support
[*] HPPD driver support
< > PHY Device support and infrastructure
< > Micrel KS8995M 5-ports 10/100 managed Ethernet switch
< > PPP (point-to-point protocol) support
< > PPP BSD-Compress compression
< > PPP Deflate compression
[*] PPP filtering
< > PPP MPPE compression (encryption)
[*] PPP multilink support
< > PPP over Ethernet
< > PPP over L2TP
< > PPP on L2TP Access Concentrator
< > PPP on PPTP Network Server
< > PPP support for async serial ports
< > PPP support for sync tty ports
< > SLIP (serial line) support
[*] SLIP compressed headers
[*] heaplative and linefill
[*] 1x bit SLIP encapsulation
< > Bluetooth Wireless LAN
[*] Wireless LAN
** Enable WMMX (Networking options) to see the WMMX drivers **
[*] WAN interfaces support
< > VMware VMXNET3 ethernet driver
[*] SON support

```

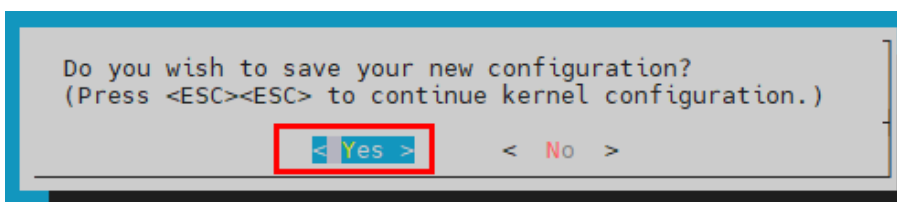
按空格键选择Multi Purpose USB Networking Framework和CDC NCM support

```

--- USB Network Adapters
< > USB CATC NetMate-based Ethernet device support
< > USB KLSI KLS0801-based ethernet device support
< > USB Pegasus/Pegasus-II based ethernet device support
< > USB RTL8150 based ethernet device support
< > Wavetek RTL8152/RTL8153 Based USB Ethernet Adapters
< > Microchip LAN78XX Based USB Ethernet Adapters
< > ** Multi Purpose USB Networking Framework **
< > ** CDC EEM support **
< > ** CDC NCM support **
< > Hsnet NCM embedded AT channel support
< > CDC NBIH support
< > avicomm OM80xx based USB 10/100 ethernet devices
< > corechip-s2 SR0700 based USB 1.1 10/100 ethernet devices
< > corechip-s2 SR0800 based USB 2.0 10/100 ethernet devices
< > MSC LAN75XX based USB 2.0 gigabit ethernet devices
< > amethyst GLEND05B-A based cables
< > Nithip 1080 based cables (Laplink, ...)
< > prolific PL-2301/2302/25A1/27A1 based cables
< > MicChip MC57830 based Ethernet adapters
< > Hst for RNDIS and ActiveSync devices
< > Simple USB Network Links (CDC Ethernet subset)
[*] ALI M5632 based 'USB 2.0 Data Link' cables
[*] anchorChips 2720 based cables (Xircom POMEET, ...)
[*] iTEK based host-to-host cables (Advance, Belkin, ...)
[*] Embedded ARM Linux Links (UPag, ...)
[*] Epson 2888 based Firmware (DEVELOPMENT)
[*] Technology KC2190 based cables (Instalnet)
< > Sharp Zaurus (stock ROM) and compatible
< > Conexant CX82310 USB ethernet port
< > Samsung Kalmis based LTE USB mode
< > GMI WMAN driver for Qualcomm MSM based 3G and LTE modems
< > Opticon USB High Speed Mobile Devices
< > intelion PLC based usb adapter
< > Apple iPhone USB Ethernet driver
< > USB-to-WMAN driver for Sierra Wireless modems
< > LG VL600 modem dongle
< > Lingheng CH9200 USB ethernet support

```

如上操作选完所需选项后，通过选择<Exit>按钮，逐层退出各个配置界面。最后在保存配置界面中，选择<Yes>选项并退出，自动更新 .config 文件。



5.3 Android 内核的 CDC ECM 驱动

5.3.1 Android 内核的编译配置

CDC ECM驱动的相关配置项，在Android_src根目录下的.config文件中，确保下面的的配置项已经被选定：

CONFIG_USB_USBNET=y

CONFIG_NETDEVICES=y

CONFIG_USB_NET_CDCETHER=y

5.3.2 Make menuconfig 开启宏配置方式

Android_src/kernel/ 目录下执行命令 “make menuconfig”

```
/Andriod9/code/kernel$ make menuconfig
```

出现Kernel Configuration界面后，进行如下配置：

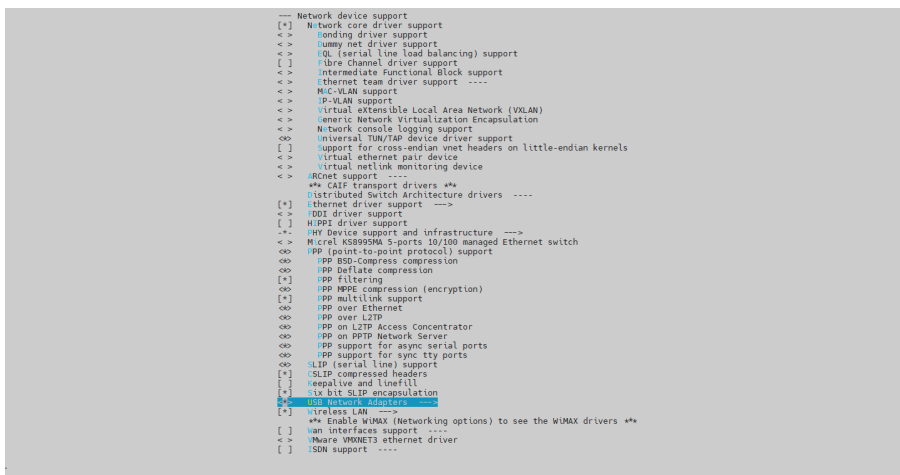
按Enter选择Device Drivers

```
--. Patch physical to virtual translations at runtime
[*] enable loadable module support --->
[*] enable the block layer --->
system type --->
bus support --->
kernel features --->
  boot options --->
  CPU Power Management --->
  floating point emulation --->
  userspace binary formats --->
  power management options --->
[*] Networking support --->
  [*] Device Drivers --->
    firmware drivers --->
    file systems --->
    kernel hacking --->
    security options --->
  --. cryptographic API --->
  library routines --->
[ ] virtualization ----
```

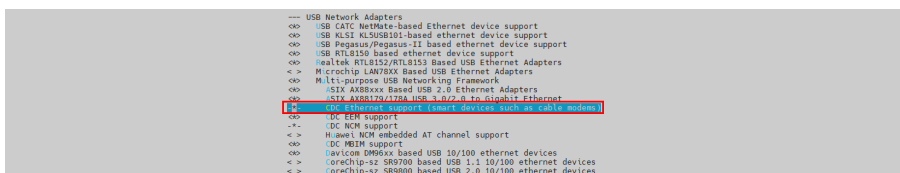
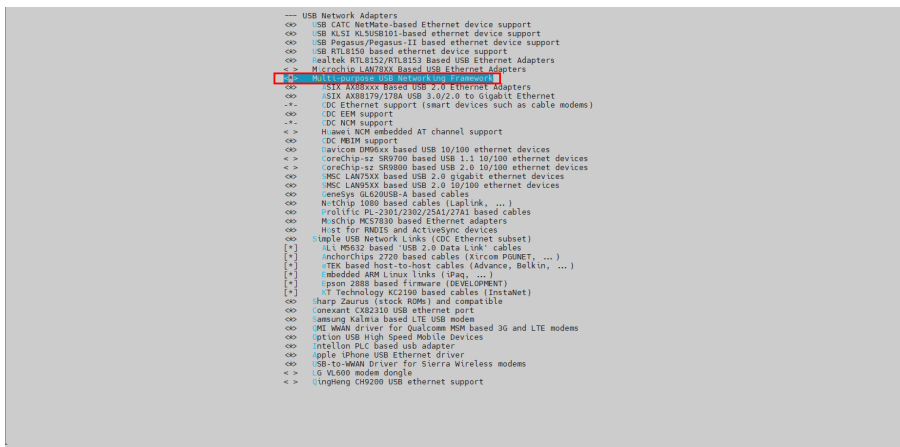
按Enter选择Network device support

```
[ ] enable AHB driver for NVIDIA Tegra SoCs
generic driver options --->
bus devices --->
< > connector - unified userspace <-> kernel space linker ----
< > M-Memory Technology Device (MTD) support ----
--. device tree and open firmware support --->
< > parallel port support ----
[*] block devices --->
  oo NVM Express block device
  Misc devices --->
  < > ATA/ATAPI/PM/RAID support (DEPRECATED) ----
  SCSI device support --->
  oo serial ATA and Parallel ATA drivers (libata) --->
  [*] Multiple devices driver support (RAID and LVM) --->
  < > generic Target Core Mod (TCM) and ConfigFS Infrastructure ----
[ ] Union HBT device support ----
  IEEE 1394 (FireWire) support --->
  [*] Network device support --->
    [ ] open-channel SSD target support ----
    input device support --->
    character devices --->
    I2C support --->
    [*] IPI support --->
    < > IPMI support ----
    < > H-II support ----
    FPS support --->
    ITP clock support --->
    I/O controllers --->
    --. IPI support --->
    < > Dallas's i-wire support ----
    --. power supply class support --->
    [*] Adaptive Voltage Scaling class support --->
    oo Hardware Monitoring support --->
    generic thermal sysfs driver --->
    [*] Watchdog Timer Support --->
    on-chip silicon backplane --->
    broadcom specific AMBA --->
    Multifunction device drivers --->
    oo Fairchild fusb302 Support --->
    --. voltage and current regulator support --->
    oo Multimedia support --->
    graphics support --->
    sound card support --->
    HID support --->
    [*] USB support --->
    < > ultra wideband devices ----
    oo MMC/SD/SDIO card support --->
    < > Sony MemoryStick card support ----
  [*]
```

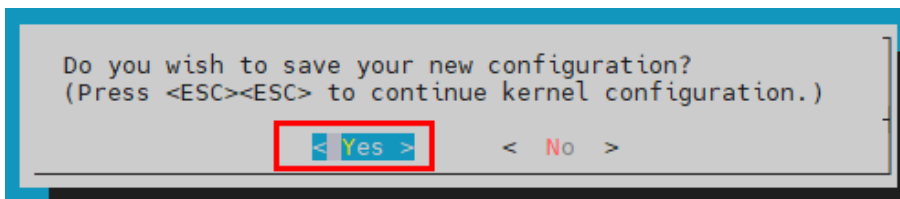
按Enter选择USB Network Adapters



按空格键选择Multi Purpose USB Networking Framework和CDC Ethernet support(smart devices such as cable modems)



如上操作选完所需选项后，通过选择<Exit>按钮，逐层退出各个配置界面。最后在保存配置界面中，选择<Yes>选项并退出，自动更新 .config 文件。



6 附录

- 6.1 如何确认系统中是否已经存在正确的USB串口驱动
- 6.2 如何确认系统中是否已经存在正确的CDC NCM驱动
- 6.3 如何获取单板当前的端口映射情况信息

6.1 如何确认系统中是否已经存在正确的 USB 串口驱动

打开Terminal，执行命令dmesg，查看内核的log信息。如查看到类似如下图内容，即说明设备正常加载。

```
134.037036] usb 1-1: new high-speed USB device number 6 using ehci-platform
134.163963] usb 1-1: config 2 interface 1 altsetting 1 has 2 endpoint descriptors, different from the interface descriptor's value: 3
134.164934] usb 1-1: New USB device found, idVendor=3466, idProduct=3301
134.164976] usb 1-1: New USB device strings: Mfr=13, Product=14, SerialNumber=15
134.165024] usb 1-1: Product: TDTECH MT571X
134.165048] usb 1-1: Manufacturer: TD Tech Ltd
134.165072] usb 1-1: SerialNumber: 0123456789ABCDEF
134.243247] cdc_ncm 1-1:2.0: MAC-Address: 0c:5b:8f:27:9a:64
134.245305] cdc_ncm 1-1:2.0 usb0: register 'cdc_ncm' at usb-fe380000.usb-1, CDC NCM, 0c:5b:8f:27:9a:64
134.247883] option 1-1:2.2: GSM modem (1-port) converter detected
134.248939] usb 1-1: GSM modem (1-port) converter now attached to ttyUSB0
134.250639] option 1-1:2.3: GSM modem (1-port) converter detected
134.251768] usb 1-1: GSM modem (1-port) converter now attached to ttyUSB1
134.253630] option 1-1:2.4: GSM modem (1-port) converter detected
134.254944] usb 1-1: GSM modem (1-port) converter now attached to ttyUSB2
134.256831] option 1-1:2.5: GSM modem (1-port) converter detected
134.258658] usb 1-1: GSM modem (1-port) converter now attached to ttyUSB3
```

6.2 如何确认系统中是否已经存在正确的 CDC NCM 驱动

打开Terminal，执行命令dmesg。如出现下图红框部分信息，则说明存在正确的CDC NCM驱动。

```
134.243247] cdc_ncm 1-1:2.0: MAC-Address: 0c:5b:8f:27:9a:64
134.245305] cdc_ncm 1-1:2.0 usb0: register 'cdc_ncm' at usb-fe380000.usb-1, CDC NCM, 0c:5b:8f:27:9a:64
```

6.3 如何获取单板当前的端口映射情况信息

(1) 执行命令dmesg，查看驱动是否加载成功。假如log信息中存在下图中红框部分信息，则说明驱动已经被成功加载（idProduct因产品的不同而不同）。

```
134.037036] usb 1-1: new high-speed USB device number 6 using ehci-platform
134.163963] usb 1-1: config 2 interface 1 altsetting 1 has 2 endpoint descriptors, different from the interface descriptor's value: 3
134.164934] usb 1-1: New USB device found, idVendor=3466, idProduct=3301
134.164976] usb 1-1: New USB device strings: Mfr=13, Product=14, SerialNumber=15
134.165024] usb 1-1: Product: TDTECH MT571X
134.165048] usb 1-1: Manufacturer: TD Tech Ltd
134.165072] usb 1-1: SerialNumber: 0123456789ABCDEF
134.243247] cdc_ncm 1-1:2.0: MAC-Address: 0c:5b:8f:27:9a:64
134.245305] cdc_ncm 1-1:2.0 usb0: register 'cdc_ncm' at usb-fe380000.usb-1, CDC NCM, 0c:5b:8f:27:9a:64
```

（2）当前模块设备的Modem和PCUI等端口的设备文件名称查询命令：ls -all /dev/ttyUSB*

```
rk3399pro:/ # ls -all /dev/ttyUSB*
crw-rw---- 1 radio radio 188, 0 2023-03-15 09:56:36.740012633 +0000 /dev/ttyUSB0
crw-rw---- 1 radio radio 188, 1 2023-03-15 09:56:36.743345967 +0000 /dev/ttyUSB1
crw-rw---- 1 radio radio 188, 2 2023-03-15 09:56:36.746679301 +0000 /dev/ttyUSB2
crw-rw---- 1 radio radio 188, 3 2023-03-15 09:56:36.750012634 +0000 /dev/ttyUSB3
rk3399pro:/ #
```

对MT571X模块，默认端口的对应关系举例如下：

Application Interface	/dev/ttyUSB0
PCUI	/dev/ttyUSB1
SerialC	/dev/ttyUSB2
GPS	/dev/ttyUSB3